



IILM UNIVERSITY, GREATER NOIDA

School of Computer Science and Engineering

SUMMER INTERNSHIP PRESENTATION

PBL Nexus

A Smart Web-Based Project & Task Management System for Project-Based Learning

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ENROLLMENT NO.

CS-2341902

PROGRAM & SEMESTER

B.Tech CSE — 7th Semester

BATCH

2023 – 2027

ORGANIZATION

Qveto Technologies

INTERNSHIP DURATION

07 Jun 2026 – 08 Sep 2026



OVERVIEW

What We'll Cover

01 About the Organization

Qveto Technologies & the internship track

02 Problem & Objectives

Why PBL Nexus was built, and its goals

03 Tech Stack & Architecture

The tools and design behind the system

04 Product Walkthrough

Login, Admin & Student experiences

05 Methodology & Outcomes

How it was built, and what it achieves

06 Certification

Proof of internship completion



Qveto Technologies

Qveto Technologies is a technology solutions and training organization offering industry-oriented internship and training programs, focused on building practical, deployable software products rather than purely theoretical assignments.

- **Domain:** Web Development
- **Certification:** ISO 9001:2015 Certified
- **Duration:** 07 Jun 2025 – 08 Sep 2025 (13 weeks)
- **Mode:** Project-Based Remote Internship

INTERNSHIP SNAPSHOT

13

Weeks of Internship

1

Full-Stack Product Shipped

6

Core Pages / Modules Built

2

User Roles Supported



THE CHALLENGE

Problem Statement

Academic institutions running Project-Based Learning programs largely rely on manual, fragmented methods — spreadsheets, emails, in-person follow-ups — to assign tasks, track progress, collect submissions and record grades.



Time-Consuming

Manual coordination eats into learning and mentoring time.



Error-Prone

Spreadsheets and email threads lose track of status and files.



No Visibility

Neither students nor instructors get real-time progress insight.



Project Objectives

1

Design a secure, role-based login system distinguishing Admin and Student users.

2

Enable admins to add users, assign tasks with due dates, and manage the workflow.

3

Allow students to view tasks, update progress, and submit files with notes.

4

Build a grading module for reviewing submissions, marks, and feedback.

5

Provide real-time dashboards summarizing task and submission status.

6

Deploy the system as a live, cloud-hosted application accessible anywhere.



Technologies & Tools Used



Frontend

HTML5 · CSS3 · Vanilla JavaScript



Backend

Node.js REST API (Express-style routes)



Deployment

Railway (cloud hosting)



Typography & UI

Playfair Display · DM Sans · dark theme



File Handling

multipart/form-data uploads via FormData API

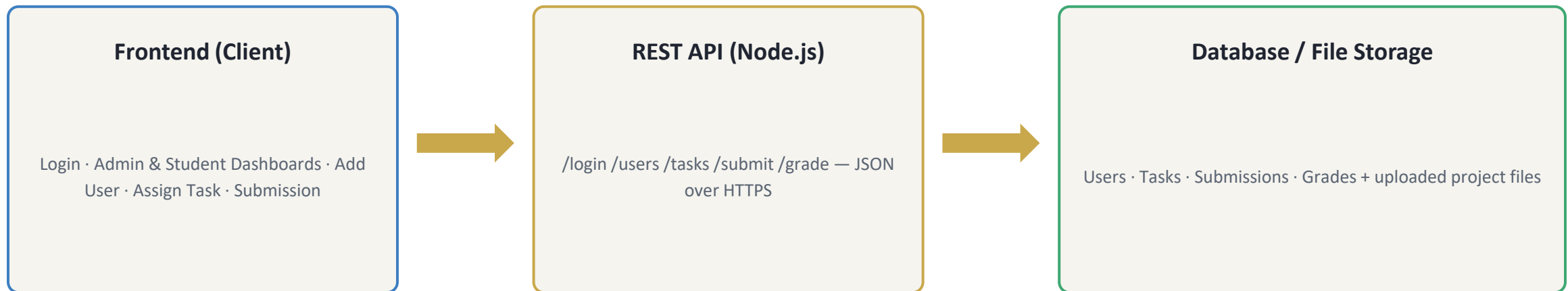


Client State

Browser localStorage for session & role routing



System Architecture



Hosted on Railway · fetch() / async-await drives every client-server exchange · role-based redirect after login (admin / student)



Secure Role-Based Login

A single entry point that authenticates users and routes them by role.



Nexus

Project-Based Learning Management System

Welcome back

Sign in to your account to continue

EMAIL ADDRESS

you@pbl.com

PASSWORD

Sign In

DEMO CREDENTIALS

Admin → admin@pbl.com / admin123

Student → alice@pbl.com / alice123

- **Unified Auth**

One login screen for Admin & Student roles.

- **Session Handling**

Role stored client-side, drives redirects.

- **Clean UI**

Dark, distraction-free academic theme.



Admin Dashboard

Full visibility and control over every student submission.

The screenshot shows the PBL Nexus Admin Dashboard. At the top, there's a header with the PBL Nexus logo and navigation buttons: ADMIN, + Assign Task, + Add User, and Logout. Below the header, the main section is titled 'Admin Dashboard' with a subtitle 'Review student submissions and assign grades'. There are three summary cards: '1 SUBMITTED' (blue), '1 REVIEWED' (orange), and '1 APPROVED' (green). Below these cards, there's a section titled 'All Submissions' with a table of student submissions.

STUDENT	TASK	FILE	NOTES	STATUS	SUBMITTED	ACTIONS
Aisha Verma	IoT Weather Station	Download	Final build attached	SUBMITTED	2025-08-10	Grade
Rohit Sharma	E-Commerce UI Redesign	Download	Figma + code	REVIEWED	2025-08-05	Graded ✓
Neha Kapoor	ML Spam Classifier	Download	92% accuracy	APPROVED	2025-07-28	Graded ✓

- **Live Stats**

Submitted / Reviewed / Approved at a glance.

- **File Access**

One-click download of student submissions.

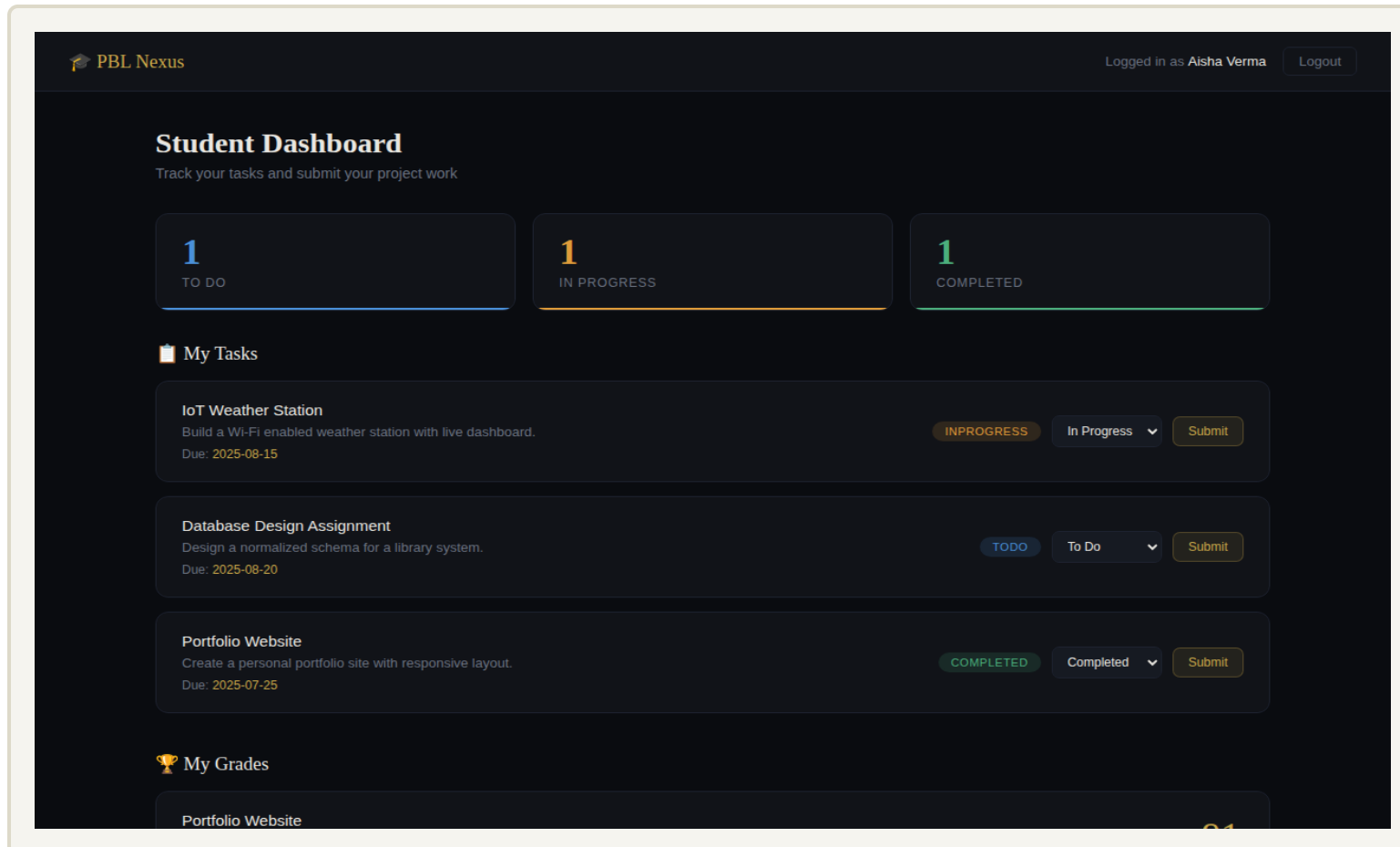
- **Grading**

Assign marks & feedback directly from the table.



Student Dashboard

A single place for students to track tasks and submit work.



- **Task Tracking**

To Do / In Progress / Completed at a glance.

- **Status Control**

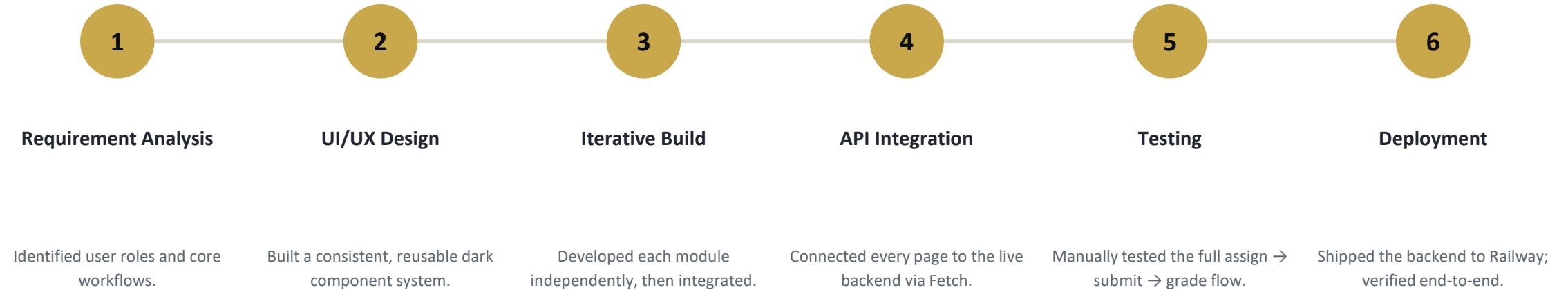
Students self-update progress in real time.

- **Grades**

Marks and instructor feedback shown per task.



Development Methodology





Expected Outcomes & Learnings

By the end of the internship, PBL Nexus was delivered as a working full-stack web application — with authentication, role-based dashboards, task tracking, file submissions, and a grading workflow with feedback.

- Frontend UI/UX design with a real design system
- Consuming REST APIs asynchronously (fetch, async/await)
- Handling file uploads (multipart/form-data)
- Role-based access & client-side session handling
- Cloud deployment of a live backend (Railway)
- End-to-end product thinking: idea → build → ship



Thank You

Questions & Discussion

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